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Research article

The nexus of carbon disclosure and green technology innovation: An empirical analysis based on Chinese listed companies

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ABSTRACT

The level of carbon information disclosure is crucial in directing the decision-making processes and activities of enterprises, particularly in the realm of green technology development. It serves a crucial function in meeting environmental standards. This research analyzes publicly traded firms on the Shanghai and Shenzhen Stock Exchanges in China from 2013 to 2020. This study used a mediating effect model to examine the influence of carbon information disclosure quality on green technology innovation. The results indicate that improving the precision of carbon data mitigates financial limitations, hence promoting innovation in green technology. The research clarifies the moderating effects in the sequence of "carbon information disclosure - financing constraints - green technology innovation," specifically about the interaction between financing constraints and green technology innovation. It illustrates that government subsidies and media focus substantially mitigate financial limitations, hence fostering innovation in green technologies, as shown by robustness tests. The study reveals discrepancies in the impact of carbon disclosure quality on the advancement of green technology, ascribed to differences in ownership, geographical variables, managerial political affiliations, and industry traits. This study enhances corporate environmental standards and provides critical insights for policymakers and business leaders, facilitating China's shift towards a sustainable, low-carbon economy.

1. Introduction

To most of the developing world, the dominant paradigm of development has been "growth first, governance later." This strategy has resulted in acute environmental problems as it has promoted fast development in industry, urbanization, and digitization. Unchecked development has led to greater ecological degradation, rising carbon emissions, global climate change, and increased severe weather events all of them posing significant threats to human lives and the sustainability of development. Faced with such issues, governments, firms, and society everywhere now place highest priority on climate governance. Corporate carbon disclosure is an inherent part of climate governance and has become increasingly significant as a voluntary instrument for improving environmental transparency. Firms not only meet rising stakeholder demands but also enhance access to green finance and investing by disclosing carbon emissions, energy efficiency practices, and sustainability commitments. Beyond legal compliance, however, by alleviating information asymmetry, enhancing investor confidence, and supporting business sustainability transformations, carbon disclosure might also spur green technology innovation. This paper intends to investigate how corporate carbon disclosure quality affects green technology innovation and to analyze the process such as finance constraints and external policy inducement that mediate this relationship.

China is now experiencing a critical phase of growth (Yang et al., 2024a). The Chinese government aims to peak carbon dioxide emissions by 2030 and cut carbon intensity per unit of GDP by 60–65 % compared to 2005 levels to address climate change. Green development has been thoroughly characterized as the optimal method for China to achieve carbon neutrality while actualizing carbon peaking. Regarding the establishment of emission reduction targets, companiesessential organizational entities within contemporary economic systems must transition to a new model, since traditional business models have become outmoded. By fostering green technological innovation, firms will not only manufacture goods that are less detrimental to the environment,

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but they will also diminish their facilities' reliance on natural resources. For instance, a primary factor is that Sinopec is a prominent oil company and petrochemical conglomerate. They have been extensively pursuing green technology experiments. Their firm has invested in research and development in renewable energy, biochemical engineering, and petroleum refining, all aimed at enhancing renewable energy and reducing carbon emissions. Sany Heavy Industry ascends to the forefront of global construction equipment manufacturers by using and developing energy-efficient and ecologically sustainable machinery solutions. Consequently, energy consumption has been markedly reduced, and critical environmental issues have been addressed.

Transitioning to a low-carbon economy, along with addressing climate change, will need green technical innovation promoted by the industry to adapt to new technological developments (Ge et al., 2025). The varied trajectory of China's migration will ultimately benefit the nation in the long term. Establishing the fundamental processes for technology transfer will advance organizations toward a more sustainable future. Measures, rules, and laws must be implemented to maintain environmental sustainability in industries. The development of corporate green technologies will be addressed via effective implementation. The main aim of green innovations is to use environmental resources efficiently, while sustainable development serves as the overarching idea (Li et al., 2024a,b,c; Yang et al., 2024b; Zhao et al., 2024a,b,c). The rise in public awareness of green energy, engagement with other green enterprises, and cooperation with many sectors all facilitate the growth of the eco-business network. It may also be seen as a method by which manufacturing conforms to a non-polluted environment. Employers may enhance their environmental sustainability via green technology innovation by reducing energy consumption, providing prompt wastewater disposal, and controlling trash through recycling. Employers are thereby reducing environmental pollution while simultaneously attaining industrial growth and structural transformation. Initiatives by industry leaders, especially those facilitating the transfer of digital and sustainable green technologies, will revolutionize industries towards greater sustainability, affecting the whole supply chain (Fatima et al., 2024).

Corporate green technical innovation is important to China's comprehensive environmental governance policy. It offers crucial assistance in attaining the nation's twin goals of peak carbon emissions and carbon neutrality, while also aiding the countrywide initiative against pollution (Qamri et al., 2022; Bingxin et al., 2025). In this framework, corporate innovation serves not only as a strategic need for firms but also as a crucial catalyst for national environmental and climatic objectives. Comprehending the processes by which green technology innovation yields environmental and economic advantages is an essential research focus with considerable ramifications for company strategy and public policy. The traditional Porter Hypothesis asserts that well structured environmental restrictions may stimulate corporate green technical innovation. The current research mostly examines this link via environmental regulation, investigating the impact of different regulatory instruments on businesses' motivations to pursue green innovation (Cao and Yu, 2024). In contrast to China's predominant strategy, which depends significantly on compulsory command-and-control environmental legislation, carbon information disclosure functions as a voluntary regulatory mechanism. Carbon disclosure enhances transparency by disseminating essential information about companies' carbon emissions, low-carbon objectives, and energy conservation strategies, while also guiding and incentivizing corporate green technological innovation (Wang et al., 2024).

This study aims to examine the impact of corporate carbon disclosure quality on green technology innovation and explore the mechanisms underlying this relationship. Specifically, we focus on:1)Assessing whether higher-quality carbon disclosure serves as a catalyst for corporate green technology innovation.2)Investigating the mediating role of financing constraints in this relationship, analyzing whether improved carbon disclosure alleviates financial limitations that hinder green innovation.3)Examining the moderating effects of government subsidies and media attention on the above mechanisms, to determine how external factors amplify or weaken the influence of carbon disclosure on innovation outco,mes.

By addressing these objectives, this research contributes to both academic discourse and policy discussions on corporate sustainability, green financing, and environmental transparency. Unlike prior studies that have predominantly focused on the role of environmental regulations in driving green innovation (Cao and Yu, 2024), this study highlights carbon disclosure as a market-driven, information-based governance tool that can shape corporate innovation strategies. Furthermore, by incorporating government subsidies and media exposure as moderating variables, we provide a more nuanced understanding of the external forces that influence the effectiveness of carbon disclosure in fostering green technological advancements.

2. Literature review and research hypothesis

2.1. Carbon disclosure and corporate green technology innovation

By increasing market transparency, decreasing information asymmetry, and improving business financial conditions, the quality of carbon information disclosure (CID) plays a most important role in facilitating green technology innovation (GTI). High-quality CID ensures timely and accurate carbon information and builds investor confidence in companies' environmental policies. Companies providing accurate and thorough carbon information fulfill regulatory standards and improve their market value by attracting Environmental, Social, and Governance (ESG)-interested investors and improving access to finance (Liu et al., 2023).

Companies with strong carbon disclosure policies show better returns on green technology investments, according to empirical research because improved openness lowers capital costs and motivates long-term R&D expenditure for low-carbon technologies (Han et al., 2024; Zhao et al., 2023). Strong governance systems, especially those of private businesses, help companies most especially when they use CID to support internal responsibility and match with market expectations led by sustainability. By proving a company's dedication to sustainability, CID's signaling impact magnifies this process even more and draws outside investment, hence driving technical advancement (Xiao et al., 2024).

Policy and regulatory wise, companies have been pushed to improve the openness of their environmental data by the worldwide tightening of carbon disclosure rules (Luo and Tang, 2023). Governments and regulatory organizations are realizing more and more CID as a market-driven tool that enhances conventional environmental regulations and motivates businesses to use low-carbon technology (Chang et al., 2023). Corporate transparency in carbon management will become a necessary instrument for obtaining regulatory approvals and competitive benefits in the green economy as global carbon reduction systems change (Xu et al., 2024).

Combined with Institutional Theory, it shows that state-owned enterprises, due to stricter government regulation, are usually more active in carbon disclosure and receive more policy support, thus promoting green technological innovation. In contrast, non-state-owned enterprises are subject to the pressure of market competition but have limited financing channels, so their carbon disclosure has a more significant effect on the alleviation of financing constraints, which further promotes green innovation.

In the eastern region, the market economy is developed and enterprises have sufficient technological reserves, so the marginal impact of carbon disclosure on green innovation is low; in the central region, which is in the period of industrial upgrading, the dual drive of policy and market makes carbon disclosure an important factor in promoting green innovation; in the western region, the infrastructure is relatively backward, but the government's support is stronger, so the carbon disclosure is mainly used as a tool to strive for government funding and

policy resources, and the impact mechanism is different from that of the influence mechanism is different from that of the east and center.Based on these findings, this study hypothesizes.

H1. The quality of carbon disclosure is a catalyst for firms' green technology innovation and is heterogeneous.

2.2. The mediating role of financing constraints

High-quality carbon information disclosure (CID) plays a crucial role in reducing financial constraints, optimizing capital structure, and fostering green technology innovation (Li et al., 2022). By enhancing market recognition of corporate sustainability efforts and improving transparency, CID mitigates investor uncertainty regarding environmental risks. This in turn widens firms' access to external finance, enabling greater expenditure on low-carbon technology R&D, energy efficiency innovation, and clean manufacturing methods (Zhao et al., 2025; Zhou et al., 2024).

Empirical evidence confirms that improving the quality of carbon disclosure reduces the costs of finance as greater transparency instills confidence in investors and promotes the financial health of a firm (Luo et al., 2019). Greater involvement by firms in environmental compliance also benefits credit ratings, decreased costs of capital, and increased attractiveness to investors who follow the use of Environmental Social and Governance (ESG) standards (Hamrouni et al., 2019; Yang et al., 2025). Strict environmental regulations reduce corporate finance limitations at a regulatory level by motivating firms to raise green innovation spending (Zhao et al., 2024a,b,c). Government-sponsored programs in carbon transparency and subsidies also improve access to low-cost finance by firms and improve their competitive advantage in sustainability industries (Tanveer et al., 2024). Quantitative carbon disclosure will play an increasingly important role in firm finance plans as carbon reduction policies develop (Xu et al., 2024).

Particularly in capital-intensive sectors, funding constraints are generally acknowledged as a significant impediment to business innovation (Li et al., 2022). Pecking Order Theory believes information asymmetry prompts firms to favor internal finance over external sources and consequently it becomes challenging for firms with funding constraints to make investments in green technology. Through improving transparency, alleviating investor uncertainty and consequently providing improved funding conditions for green projects, higher CID offsets funding constraints (Zhou et al., 2024).

Also, trustworthy carbon reporting, according to Signaling Theory, signals financial markets in a positive manner and consequently improves the creditworthiness of companies and the cost of capital (Ren et al., 2024; Xu et al., 2024). Better access to finance allows companies to formulate long-term innovation plans and consequently cause sustainable technological development. These results guide this study to hypothesize.

H2. Enhancing the quality of corporate carbon disclosure can promote green technology innovation by alleviating financial constraints.

2.3. The moderating effect of government subsidies

Particularly in green technology development, government subsidies have an extremely important role in relieving financial pressures and increasing business innovation capability (Ji and Miao, 2020). Subsidy reduces the expenditure on R&D, promotes low-carbon technology investment, and promotes the process of shifting towards sustainable production patterns by providing direct financial support, fiscal incentives, and policy support (Wang et al., 2023). This financial support not only relieves the pressure of high-risk innovation but also helps firms to cope with market fluctuations and regulatory pressures and thus encourages their green innovation commitment in the long term (Li et al., 2024a,b).

It is predominantly the government incentives that are responsible for green innovation and carbon disclosure complementarity. Both a financial facilitator and a strategic guide, subsidies allow companies to allocate resources to low-carbon technology and energy-efficient solutions (Agrawal et al., 2024; Hunjra et al., 2024). Government incentives enable companies to balance between technological development in the long run and profitability in the short run by compensating for resource constraints and augmenting exploratory innovation and consequently making sustainable innovation viable (Lyu et al., 2024).

In addition, government support encourages programs of carbon transparency. Targeted financial support will most likely be secured by those firms actively reporting their carbon emissions because transparency fits into national sustainability goals and regulatory frameworks (Lyu et al., 2024). Companies raise their market value by means of attractive refinancing circumstances and policy-driven incentives, therefore fostering development in green technology innovation concurrently. This mutually reinforcing connection emphasizes how government subsidies help to improve the link between carbon disclosure and company innovation, hence improving companies' competitive edge in the transition to a green economy. These results lead this work to hypothesize.

H3. Government subsidies influence the extent to which financing constraints mediate the relationship between corporate carbon disclosure quality and green technology innovation.

2.4. The moderating effect of media attention

Corporate environmental openness is shaped in great part by media, which also affects investor attitude, regulatory control, and green technology development. Media coverage of companies' environmental performance helps to build public confidence and trust by means of which a company's sustainability commitment is reinforced and its long-term development prospects are improved (Khalid et al., 2024). Higher media awareness companies so usually function in a more favorable financial climate, which improves their access to resources required for research and development on green technologies.Media serves as an outside governance tool that forces companies to improve carbon disclosure criteria and match with sustainability criteria (Pan et al., 2024). Transparency itself motivates businesses to embrace green innovation projects, therefore guaranteeing environmental compliance and strengthening company image. Particularly with regard to environmental mismanagement, negative media coverage raises society and regulatory pressures, which typically results in more stringent compliance rules, more expenditures in green R&D, and changes in company sustainability plans (Tyler et al., 2024).

From the perspective of the capital market, media reporting substantially lowers the cost of finance and risk of investments. Good environmental reporting attracts ESG investors and consequently promotes companies' access to sustainable sources of finance (Chen et al., 2022). Negative environmental reporting, conversely, will destroy market confidence and lead to government fines and capital outflows (Hao et al., 2025). Companies actively reporting on carbon and integrating sustainability into business plans will likely enjoy investor support, avoid capital expenditure and improve financial conditions for green innovation (Chen et al., 2023).

Exceeding financial benefits, media accountability affects business strategic development and decision-making. Companies subject to higher media scrutiny are likely to emphasize environmental activities, carbon reporting, and investments in green technology in an attempt to diminish risk and improve market competitiveness (Bataineh et al., 2024). This linked relationship between media scrutiny, corporate transparency, and green innovation points out the major role of media in hastening companies' shift towards a sustainable business strategy. These results lead this study to hypothesize.

H4. Media attention plays a regulatory role in how financing constraints mediate the relationship between the quality of corporate

carbon disclosure and green technology innovation.

3. Research design

3.1. Variable definition

3.1.1. Dependent variable: green technology innovation

Previous studies utilizing various methods have extensively discussed the measurement of green technological innovation (GTI). Previous studies generally use two major methods: As corporate ecological innovation indicators (Shen et al., 2021;Lv et al., 2021; Liu et al., 2023), this method follows technological progress in waste management, material use sustainability, energy efficiency, and emission reduction programs. Although this method offers a qualitative view of green innovation, its limitations are generally constrained by the standardization of technological progress between industries. Another widely recognized approach lies in the number and success of green patent applications, as a purpose and measurable indicator of a firm's intention to create green-friendly innovations (Su et al., 2023). High comparability and validity in patent data make it a good proxy for green technology progress.

Green patent filings are utilised in this study as the key indicator of corporate green technology innovation based on the strengths of patent-based indicators in providing objectivity, replicability, and comparability across industries. For skewness and outlier problems in the data specifically, the logarithmic transformation of one plus green patent filings are utilised (Song et al., 2023;Yu et al., 2023). Robustness tests are made with the utilisation of the logarithm of one plus awarded green patents in order to test this measurement further and obtain consistent results across multiple patent-based indicators.

3.1.2. Explanatory variable: carbon disclosure quality

This work guarantees an objective and systematic evaluation by combining text analysis techniques with entropy weighting techniques to estimate corporate carbon disclosure quality (CID). Textual analysis is commonly used in financial and economic literature and is carried out through three straightforward methods: the bag-of-words (BoW) approach with words represented as vectors to facilitate quantitative comparisons of firms' disclosure content; dictionary-based analysis based on statistical frequency-based word analysis to identify predefined carbon-related words; and word embedding to capture semantic patterns, tone and level of disclosure. Based on the indicator approach bySong et al., (2024), we construct a weighted CID based on leading features from environmental disclosures, annual reports, and firm-specific ESG filings. Strategic commitment, governance structure, engagement and performance risk management, green innovation investments, and third-party assurance are such features. Our work applies the entropy weighting methodology, where objective weights are assigned to various disclosure elements based on between-firm variability and thus improving measurement accuracy and consequently minimizing subjectivity in the evaluation. Different validation procedures contribute to ensuring the solidity and quality of our CID index as follows: data source validation based on authoritative corporate and ESG reports; consistency tests between our index and external carbon disclosure grades; and sensitivity tests on various weighting methods and on various sets of keywords. Through such methodology improvements, our evaluation of carbon disclosure quality is transparent, replicable and in accordance with highest standards in corporate sustainability literature and thus providing a scientifically solid basis for measuring firms' environmental openness.

3.1.3. Mediating variable: financing constraints

The present work adopts an established statistic in corporate finance literature—the SA index—in an effort to capture funding limitations. Financing limitations refer to the limitations companies have in obtaining external funds, and such limitations might stifle investments in innovation and green technology development. Different indicators have been used in prior studies in an effort to capture financial limitations including the SA index, KZ index, and WW index (Tanveer et al., 2024; Wang et al., 2021). Of them, the SA index is adopted because of its strong theoretical basis, capability in handling endogeneity issues, and broad industry adoption (Lin and Ma, 2022) The SA index can be computed by the following formula:$SA = 0.737SIZE + 0.043SIZE^2 - 0.04AGE$,where SIZE is the natural logarithm of total assets (million) and AGE is the age of the firm in years. The SA index by nature is negative; accordingly, we use its absolute value in an effort to ease interpretation; larger absolute values represent tighter funding limitations.

3.1.4. Moderating variables

(1) **Media Coverage:** We use the approach used in previous research (He et al., 2024) which assesses a company's exposure in both conventional and digital media to evaluate media coverage. Media coverage is a major outside effect on company conduct as it shapes public opinion, investor mood, and regulatory scrutiny on environmental policies. Our statistic is based on the total number of news items and reports referencing the firm, gathered from both diggersources (online news platforms, social media, and financial websites) and traditional (print newspapers, financial reports, and industry magazines). Using the following method to correct for skewness and lessen the influence of extreme values, log-transformed final media exposure measure is Media Coverage = ln(1+ Total Mentions), in where "Total Mentions" is the combined count of firm-related media stories. Including 1 before the logarithm transformation guarantees that companies with 0 mediamentions do not get unclear values.

(2) **Government Grants:** Key financial incentive mechanism government offers direct funding for company sustainability projects and innovation, therefore facilitating Based on past research (Zhao et al., 2024a,b,c; Adebayo and Ullah, 2023), we assess government financial assistance by aggregating the whole government subsidy amount acquired by a company during a particular fiscal year. This statistic shows the degree of financial support legislators provide companies for sustainability changes and green technologies investment.

We use a logarithmic modification to normalize the distribution and reduce the impact of extreme values considering the significant diversity in subsidy levels across businesses and sectors. The measuring recipe is as follows: Government Grants = ln(1+Total Government Subsidies), where "Total Government Subsidies" is the overall financial assistance a company gets from local or national government programs. Including 1 before log transformation guarantees that companies without any subsidies are suitably represented in the dataset without producing arbitrary numbers.

3.1.5. Control variables

This study integrates many control variables into the model, building on prior studies and recognizing the influence of environmental information disclosure quality on company competitiveness. Factors include the firm's size, age, growth rate, equity ownership concentration, audit quality, board size, and the percentage of shares owned by institutional investors. Comprehensive descriptions of each variable are included in Table 1.

3.2. Sample selection and data sources

This study analyzes data from companies listed on the Shanghai and Shenzhen stock exchanges between 2013 and 2020, using certain selection criteria. Financial and insurance companies were excluded, along with firms that had ST or PT designations, those that were delisted

Table 1
Variable definition.

Variable Type	Name	Symbol	Definition
Dependent variable	Green technological innovation	GINV	Take the natural logarithm after adding 1 to the number of green patent applications by enterprises
Explanatory variables	Quality of carbon information disclosure	CID	Obtained through text analysis and entropy weight method
Mediating variables	financing constraints	SA	Referring to the practices of Wu and Huang
Moderator Variable	government subsidies	GOV	The natural logarithm of the total amount of government subsidies
	Media attention	MED	Take the natural logarithm after adding 1 to the sum of the times a company is reported on in online and print media
Control Variable	Asset liability ratio	LEV	Total liabilities at the end of the year divided by total assets at the end of the year
	Cash flow ratio	CASH	Net cash flow generated from operating activities divided by total asset
	ownership concentration	TOP	The proportion of shares held by the largest shareholder
	Independent director ratio	INDEP	Independent directors divided by the number of directors
	Institutional shareholding ratio	INST	Institutional shareholding ratio
	Industrydummy variables	IND	Industry dummy variables
	Year dummy variable	YEAR	Year dummy variable

or newly listed during this period, and those displaying data anomalies. The assessment of carbon information disclosure quality was meticulously obtained from annual reports, corporate social responsibility reports, and environmental reports of publicly traded companies. Data on green technology innovation was sourced from the CNRDS, focusing on a database that tracks green patents held by these companies. Information on media coverage was obtained from the "China Important Newspaper Full-text Database" under the CNKI. Additional data were acquired from the Guotai An and Wind databases. The data preparation and subsequent empirical analysis were conducted using Stata 17.0 software.

3.3. Model design

This study designs a model with GINV as the dependent variable and CID as the independent variable, incorporating industry fixed effects and year fixed effects. The model is specified as:

$$\mathrm{GINV}_{i,t} = \alpha_0 + \alpha_1 \mathrm{CID}_{\ i,t-1} + \alpha_3 \mathrm{Controls}_{\ i,t} + \epsilon_{i,t}$$

In this section: i and t represent locations and years, $\mathrm{GINV}_{i,t}$ is the dependent variable for green technology innovation of the enterprise; $\mathrm{GINV}_{i,t-1}$ is the explanatory variable for carbon information disclosure quality, $\mathrm{Controls}_{\ i,t}$ represents the control variables, and $\epsilon_{i,t}$ is the random disturbance term.

The main objective of this study is to explore how carbon information disclosure (CID) affects corporate green technology innovation (GTI) through the mediating variable of financial constraints. Based on the classic approach of Baron and Kenny (1986)and MacKinnon et al., (2007) on the mediation effect test, we believe that the mediation effect model can clearly reveal the causal paths between the variables and quantify the magnitude of the mediator's effect. Since the core of this study is to identify and quantify the mediating role of financing constraints between CID and GTI rather than constructing complex latent variable relationships, the classic mediation effect model is chosen to be more operational and explanatory.

To verify H1 and H2, this article adopts the mediating effect test method proposed by Wen and Ye, considering the lag effect of corporate carbon information disclosure. The following regression model is specified:

$$\mathrm{GINV}_{i,t} = \alpha_0 + \alpha_1 \mathrm{CID}_{\ i,t} + \alpha_2 \mathrm{Controls}_{\ i,t} + \epsilon_{i,t}$$

$$\mathrm{SA}_{\ i,t} = \beta_0 + \beta_1 \mathrm{CID}_{\ i,t} + \beta_2 \mathrm{controls}_{\ i,t} + \epsilon_{i,t}$$

$$\mathrm{GINV}_{\ \mathrm{i,t}} = \gamma_0 + \gamma_1 \mathrm{CID}_{\ \mathrm{i,t}} + \gamma_2 \mathrm{SA}_{\ \mathrm{i,t}} + \gamma_3 \mathrm{Controls}_{\ \mathrm{i,t}} + \epsilon_{i,t}$$

In this equation: $\mathrm{SA}_{\ i,t}$ is the mediating variable representing financial constraints, and the remaining variables are defined as described above.

To verify H3 and H4, this paper adopts the moderated mediation effect test method proposed by Wen and Ye, (2014) (moderated mediation through the mediated half-path), and constructs the following model to test the moderating effects of government subsidies and media attention.

$$\mathrm{GINV}_{\ i,t} = a_0 + a_1 \mathrm{CID}_{i,t} + a_2 M_{i,t} + a_3 \mathrm{Cid}_{\ i,t-1} M_{i,t} + a_4 \mathrm{Controls}_{\ i,t} + \epsilon_{i,t}$$

$$\mathrm{SA}_{\ i,t} = b_0 + b_1 \mathrm{CID}_{\ i,t} + b_2 M_{i,t} + b_3 \mathrm{Cid}_{\ i,t} M_{i,t} + b_4 \mathrm{Controls}_{\ i,t} + \epsilon_{i,t}$$

$$\mathrm{GINV}_{\ i,t} = c_0 + c_1 \mathrm{CID}_{i,t} + c_2 M_{i,t} + c_3 \mathrm{Cid}_{\ i,t-1} M_{i,t} + c_4 \mathrm{SA}_{\ i,t} + c_5 \mathrm{SA}_{\ i,t} M_{i,t} + c_6 \mathrm{Controls}_{\ i,t} + \epsilon_{i,t}$$

In the above paragraph, $M_{i,t}$ denotes the moderating variable. $\mathrm{Cid}_{\ i,t} M_{i,t}$ denotes the interaction term between carbon information disclosure and the moderating variable, whereas $\mathrm{SA}_{\ i,t} M_{i,t}$ signifies the interaction term between financial constraints and the moderating variable. $\mathrm{Controls}_{\ i,t}$ denotes the control variables, whereas $\epsilon_{i,t}$ signifies the disturbance term.

Table 2 presents the descriptive statistics for the relevant variables. The mean of green technology GINV is 0.188, with a high of 4.007 and a minimum of 0, indicating substantial heterogeneity in green technology innovation across the evaluated enterprises. The average quality of corporate carbon information disclosure (CID) is 0.062, with a range from 0 to 0.445, indicating that most enterprises have not disclosed carbon information, which reflects a deficient understanding of carbon information disclosure.

4. Empirical analysis results

4.1. Baseline regression and mediation analysis

The baseline regression results are presented in Table 3, which suggests a significant positive correlation between company green technology innovation and CID. In column 1, the regression coefficient for CID is 0.5061, which is statistically significant at the 1 % level when green technological innovation is used as the dependent variable. This research posits that the disclosure of superior carbon information is correlated with increased business involvement in the development of green technologies, thereby validating Hypothesis H1.

The fundamental premise is that enhanced carbon disclosure is

Table 2
Descriptive statistics of variables.

Variable	N	Mean	SD	Min	Max
GINV	5285	0.188	0.488	0	4.007
CID	5285	0.0620	0.0590	0	0.445
Lev	5285	0.490	0.203	0.008	2.290
Cashflow	5285	0.0530	0.0690	−0.483	0.523
Top1	5285	0.361	0.158	0.0300	0.900
INST	5285	0.494	0.226	0	1.870
Indep	5285	0.376	0.0570	0.182	0.800
Board	5285	2.177	0.208	1.386	2.890

Table 3
Quality of carbon information disclosure, financing constraints, and green technology innovation.

	(1)	(2)	(3)
	GINV	SA	GINV
CID	0.5061***	−0.1692**	0.4897***
	(0.1356)	(0.0558)	(0.1356)
Lev	−0.1522**	−0.0173	−0.1529**
	(0.0719)	(0.0295)	(0.0718)
Cashflow	−0.2403**	−0.0268	−0.2375**
	(0.1129)	(0.0464)	(0.1127)
Top1	−0.1080	−0.0065	−0.1101
	(0.1421)	(0.0584)	(0.1419)
INST	0.0314	−0.0471**	0.0244
	(0.0582)	(0.0239)	(0.0582)
Indep	−0.2866	0.2964***	−0.2505
	(0.1996)	(0.0821)	(0.1997)
Board	−0.0975	−0.0173	−0.1018
	(0.0720)	(0.0296)	(0.0719)
SA			−0.0833**
			(0.0379)
_cons	0.5876**	−3.8394***	0.2711
	(0.2188)	(0.0899)	(0.2624)
IND/YEAR	YES	YES	YES
R^2	0.474	0.678	0.475

Note: The values in parentheses are t-statistics. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

indicative of a company's dedication to environmental accountability and transparency, which in turn demonstrates a greater dedication to the mitigation of carbon emissions and sustainability concerns. Companies that possess exceptional disclosure policies are more likely to allocate significant resources to research and development (R&D) for green technology, thereby encouraging innovation that is directed toward reducing environmental impact. This strategic connection with sustainable objectives enhances the business's image, thereby gaining the confidence of investors, the support of consumers, and the recognition of a broader range of stakeholders.

Column (2) investigates how CID could help to lower business finance limitations; the regression coefficient of −0.1692 is significant at the 5 % level. This finding implies that generally speaking, companies which show better carbon disclosure have far less financial restrictions. Reducing information asymmetry and improving carbon reporting transparency helps to build confidence between lenders and investors, therefore enabling wider access to more reasonably priced equity financing and debt. Reducing financial restrictions helps companies to speed their switch to low-carbon business models, thereby advancing green technologies and strengthening their will to support sustainable development. Column (3) additionally incorporates as a mediator financial limitations (SA). At the 1 % level, the regression analysis shows that the CID coefficient is 0.4897 and the SA coefficient is −0.0833 both of which are statistically significant. The significantly positive coefficient for CID in column (1) and the significantly product of the SA coefficients in columns (2) and (3) help to establish the partly mediating influence of financing constraints on the connection between CID and green technology innovation. This study shows that by eliminating financial obstacles, the publication of carbon information both directly promotes green innovation and indirectly stimulates invention.

The method by which excellent carbon disclosure supports green innovation functions and reduces financing constraints across many channels is clear-cut. Increased carbon transparency ultimately helps creditors and shareholders to more accurately evaluate a company's environmental risks, therefore lowering financing costs and increasing investor trust. Organizations with less financial restrictions are therefore more free to devote resources to projects including green research and development as well as innovation. The fact that companies improve their competitive advantage in the green economy by investing funds to the creation of sustainable goods and cleaner technologies supports hypothesis H2. These results highlight the important role carbon disclosure laws play in improving company financial situation, encouraging technical innovation, and enabling the change to a sustainable economic model.

4.2. Analysis of the moderating effect

The empirical results of Table 4 illustrate the moderating effect of government subsidies on the relationship between SA, GINV, and CID. The coefficient for the interaction term between government subsidies and CID is not statistically significant, as indicated in Column 1. This suggests that government subsidies do not have a substantial moderating effect on the direct relationship between CID and GINV. This outcome may be attributed to the limited impact of governmental financial assistance on GINV, or to the disparate effects that subsidies have across diverse business sectors or at different stages of technical advancement.

In column (2), the coefficient of CID on SA is −0.1666, which is statistically significant at the 1 % level. This demonstrates that CID significantly alleviates finance constraints. Enhanced carbon transparency reduces information asymmetry, thereby enhancing the trust of investors and creditors. This, in turn, facilitates the acquisition of external funding under more favorable terms.

Column (3) conducts additional research on the relationship between GOV and SA, revealing a coefficient of −0.0202 that is statistically significant at the 5 % level. This conclusion suggests that an increase in GOV alleviates the financial constraints faced by enterprises, thereby indirectly promoting GINV. In the latter phase of the financing process, the moderating influence of GOV is particularly noteworthy, as financial assistance enables enterprises to overcome resource constraints and participate in sustainable technological advancement.

These findings indicate that government subsidies do not directly influence the relationship between CID and GINV; nevertheless, they indirectly promote green technological breakthroughs by alleviating financial limitations. This substantiates the legitimacy of the moderated mediation model, indicating that enterprises getting more government financial assistance encounter a less adverse effect of funding limitations on GINV. The results empirically validate Hypothesis H3, supporting the idea that targeted government subsidies might facilitate business sustainability transitions by enhancing financial accessibility for green activities.

Table 5 analyzes the role of MED as a moderating variable in the link among the quality of CID, SA, and GINV. The findings in column (1) demonstrate that the interaction term between CID and MED lacks

Table 4
Analysis of the regulatory effect of government subsidies.

	(1)	(2)	(3)
	GINV	SA	GINV
CID	0.5138***	−0.1666**	0.4935***
	(0.1356)	(0.0557)	(0.1356)
GOV	0.0085**	−0.0045**	0.0083**
	(0.0035)	(0.0014)	(0.0035)
CID*GOV	−0.0662	−0.0463**	−0.0668
	(0.0563)	(0.0232)	(0.0564)
SA			−0.0801**
			(0.0379)
SA*GOV			−0.0202**
			(0.0139)
_cons	0.4140*	−3.7584***	0.1170
	(0.2288)	(0.0940)	(0.2693)
CONTROLS	Y	Y	Y
IND/YEAR	Y	Y	Y
R^2	0.475	0.679	0.477

Table 5
The moderating effect of media attention.

	(1)	(2)	(3)
	GINV	SA	GINV
CID	0.5050***	−0.1785**	0.4917***
	(0.1359)	(0.0558)	(0.1359)
MED	0.0127**	−0.0032	0.0118*
	(0.0062)	(0.0025)	(0.0062)
CID*MED	−0.0112	0.1010**	0.0027
	(0.0905)	(0.0372)	(0.0904)
SA			−0.0824**
			(0.0379)
MED*SA			−0.0462**
			(0.0229)
_cons	0.5647**	−3.8279***	0.2361
	(0.2191)	(0.0900)	(0.2626)
CONTROLS	Y	Y	Y
IND/YEAR	Y	Y	Y
R^2	0.475	0.678	0.476

statistical significance, indicating that media exposure does not directly affect the correlation between CID and GINV. The impact may be indirect, suggesting that while MED influences public perception and business reputation, it does not substantially change the direct effect of CID on scientific progress.

The coefficient of CID on SA in column (2) is −0.1785, which is statistically significant at the 1 % level. This indicates that CID significantly mitigates funding obstacles. Enhanced transparency in environmental reporting fosters greater investor and lender confidence, thereby facilitating the acquisition of funding under more favorable terms for enterprises. The interaction between SA and MED is further investigated in Column (3), which yields a coefficient of −0.0462, which is statistically significant at the 5 % level.

This suggests that enterprises' financial status is positively influenced by improved media coverage, thereby alleviating capital constraints. Enterprises are granted improved access to funds and resources as a result of reduced financial constraints, which in turn encourages a greater investment in the research and development of green technologies. This financial adaptability allows the company to respond to changing market requirements and sustainability mandates, thereby accelerating the transition to enduring environmental accountability and low-carbon innovation.

Our findings together confirm that media attention serves as a significant indirect moderating factor in the financing process. Media criticism often affects how financial constraints drive company investment in sustainable technology, rather than directly altering the relationship between CID and green innovation. The influence is particularly pronounced in the latter stages of the process, since increased public exposure helps companies mitigate the adverse effects of financial constraints on innovation efforts. The results experimentally confirm Hypothesis H4, emphasizing the crucial role of media attention in promoting business sustainability and advancing green technology development.

4.3. Robustness tests

4.3.1. Instrumental variable method

This research utilizes an instrumental variable (IV) strategy, as outlined by Singh et al. (2023) to mitigate possible endogeneity issues stemming from omitted variable bias or reverse causation. The two-period lag of the explanatory variable is designated as the instrument. Employing delayed values mitigates simultaneity bias and improves the robustness of causal inference, since historical values act as exogenous predictors of present explanatory variables while being less affected by contemporaneous disruptions.

The first-stage regression results, shown in column (1) of Table 6, reveal an instrumental variable coefficient of 0.408, significant at the 1 % threshold. The regression reveals a R^2 value of 0.2169 and a first-stage F-statistic of 799.306, much beyond the threshold of 10, thereby disproving the null hypothesis of a weak instrument. The results confirm that the chosen instrumental variable is both legitimate and significantly associated with the endogenous explanatory variable. The results of the second-stage regression, shown in column (2) of Table 6, further validate the model's robustness. The findings indicate that the quality of corporate carbon information disclosure has a considerable beneficial effect on green technological innovation, reinforcing the idea that enhanced transparency in carbon reporting encourages corporate engagement in sustainable technology advancement.

4.3.2. Altering the mediating variables

This research utilizes the KZ index as an alternative mediating variable to assess the robustness of the results, considering the variety of metrics for analyzing corporate finance limitations. The mediation study indicates that, irrespective of the KZ index or the SA index used, the regression coefficient of CID remains considerably negative. This outcome corroborates other research, reaffirming the idea that superior carbon disclosure significantly alleviates financing limitations by improving business transparency and bolstering investor trust.

In column (3) of the regression findings, CID demonstrates a considerably positive coefficient, whereas finance restrictions reveal a significantly negative coefficient. Furthermore, the product of the coefficients for finance constraints in columns (2) and (3) is positive, indicating that the mediating effect of funding restrictions remains intact despite changes to the mediating variable. These findings further substantiate that the quality of corporate carbon disclosure directly impacts green technology innovation and indirectly promotes it by alleviating financial obstacles. The uniformity of these results across various mediating variable configurations validates the robustness of the mediation effect and enhances the dependability of the suggested theoretical framework.

4.3.3. Changing the dependent variable

Several approaches are used in this work to evaluate the dependent variable, therefore guaranteeing the validity of the results. Introduced to indicate whether a company has sought green technology patents in a given year is a binary (0–1) variable. The first conclusions are confirmed by the considerable positive association between the quality of CID and GINV shown by the regression results in column (4) of Table 7. Second, the advancement of green technologies is evaluated using another yardstick. Since this research more fairly represents the technical developments attained by companies, it evaluates the number of approved green patents rather than patent applications. Table 7's Column (5) shows that the regression results consistently show a substantial positive connection between CID quality and GINV, therefore strengthening the validity of the conclusions. Strong empirical evidence that the link

Table 6
Instrumental variable method.

Variables	(1)	(2)
	First Stage	Second Stage
IV	0.4080***	
	(28.27)	
CID		1.0781***
		(3.34)
_Cons	0.1331	0.9422
	(0.87)	(0.59)
Controls	Y	Y
Weak instrumental variable	799.306	
Underidentification test	638.909	
R^2	0.2169	0.0314

Table 7
Robustness test.

	(1)	(2)	(3)	(4)	(5)
	GINV	KZ	GINV	GINV (0–1)	GINV(A)
CID	0.5061***	−1.4273**	0.4624**	0.2427**	0.6852***
	(0.1356)	(0.6536)	(0.1424)	(0.1090)	(0.1446)
KZ			−0.0074**		
			(0.0035)		
_cons	0.5876**	2.9503**	0.6204**	0.4917**	0.3293
	(0.2188)	(1.0357)	(0.2258)	(0.1758)	(0.2333)
CONTROLS	Y	Y	Y	Y	Y
IND/YEAR	Y	Y	Y	Y	Y
R^2	0.474	0.378	0.477	0.424	0.463

between CID and GINV is both robust and stable, independent of measurement differences, comes from the consistency of these findings across many definitions of dependent variables.

5. Further discussion

5.1. Corporate equity characteristics and regional test

This study conducts distinct regression analyses for SOEs and NSOEs to investigate the varied effects of corporate ownership structure on the relationship between CID and green technology innovation. The regression findings in column (1) demonstrate a substantial positive link between CID quality and green technology innovation in SOEs. This indicates that state-owned enterprises, often subject to more rigorous regulatory scrutiny and policy directives, gain from carbon disclosure via better environmental governance and augmented technical advancement. Column (2) indicates that the beneficial effect of CID on green technology innovation is much more prominent in NSOEs.

Subject to tighter government scrutiny and environmental regulation on a frequent basis, state-owned enterprises (SOEs) have to fulfill national sustainable development targets. Even though it ensures firms achieve legislative requirements and promotes their environmental governance, quality carbon disclosure also allows SOEs to project social responsibility and transparency. In response to fulfilling the government's low-carbon development strategy, SOEs have been pushed to develop aggressively in green technology by institutional pressure. As SOEs contribute significantly to the country's economic development strategy, the quality of carbon disclosure could send good policy signals to the government and the market and thus show the commitment of the enterprises in fulfilling their environmental responsibilities. This signaling mechanism could allow institutional investors, government authorities, and overseas markets to gain confidence and thus support the role of SOEs in green development.

Unlike SOEs, NSOEs operate mainly in a more competitive market environment, where their strategic decisions are more influenced by market demand, financing constraints and investor preferences. NSOEs usually face greater financing constraints and find it difficult to support capital-intensive green technological innovations through internal funds. Therefore, by increasing the transparency of carbon disclosure, NSOEs can reduce information asymmetry and enhance investor confidence, thereby increasing the likelihood of obtaining bank loans, bond financing and equity investment. Since investors usually perceive non-SOEs as having higher investment risks, their market valuation tends to be lower, making it more difficult to obtain low-cost financing. Unlike policy-driven SOEs, non-SOEs need to proactively build competitive advantages in the marketplace to attract capital, customers and business partners.

This study uses a separate regression analysis for businesses in the east, middle, and west areas to explore the regional differences in the relationship between CID and green technology innovation (columns (3)–(5) of Table 8). The results of column (3) illustrate that the effect of the green technology innovation is stronger in the companies located in the central area compared to the ones in the western region not to mention those in the eastern region which is negligible.

The eastern region is the most economically developed and marketized region in China, with a well-developed industrial base, a high level of technological innovation capacity and abundant capital market resources. Therefore, enterprises in this region have already established a more mature system in green technology innovation, and the marginal promotion effect of carbon information disclosure is relatively small. The eastern region has a large number of high-tech enterprises, R&D institutions and internationalized enterprises, and the technology spillover effect among these enterprises is obvious, which makes the diffusion of green technology innovation more likely to occur. As companies already have more mature environmental technologies and industrial synergies, green innovation is more market-driven rather than relying on carbon disclosure as a driving factor. As investors have already built up confidence in the green technology innovation capability of enterprises, the marginal contribution of carbon disclosure to financing is relatively low, and therefore further improving the transparency of carbon disclosure has a more limited role in promoting technological innovation.

Compared with the eastern region, the central region is at a critical stage of rapid economic development and industrial restructuring, so carbon information disclosure plays a more significant role in promoting green technological innovation. Driven by the government, industries in the central region are transforming from traditional manufacturing to high-tech, low-carbon and environmentally-friendly industries, and there is a strong demand for technological innovation. Companies must show their environmental commitment by carbon disclosure as the federal government has tightened its control of business environmental compliance and implemented tougher carbon emission management laws in recent years. Apart from a need for businesses to get government subsidies and tax benefits, carbon disclosure has evolved into a crucial instrument for them to improve their industry image and market competitiveness.

The western region is relatively economically backward and has a weak industrial base, but in recent years, green development and sustainable economic transformation have become an important development direction under the promotion of the government, and the impact of carbon information disclosure on green innovation has gradually increased. Western governments encourage enterprises to develop green technologies through financial subsidies, special funds and low-interest loans, and at the same time require enterprises to improve the transparency of carbon disclosure in order to obtain policy support. Carbon disclosure is not only a necessary condition for enterprises to obtain government resources, but also a way for enterprises to demonstrate their commitment to sustainable development and enhance market recognition. As the capital market is still immature, enterprises face higher credit risks and stricter loan approval standards in financing.

Table 8
Heterogeneity Test:Corporate equity characteristics and regional test.

	(1)	(2)	(3)	(4)	(5)
	N-SOE	SOE	EAST	MID	WEST
CID	0.5687***	0.3236***	0.3382**	1.3968***	0.3218
	(0.1548)	(0.2487)	(0.1583)	(0.3827)	(0.3469)
_cons	0.6616**	0.3557	0.5381**	0.4484	0.3801
	(0.2600)	(0.3888)	(0.2702)	(0.6098)	(0.4837)
CONTRONLS	Y	Y	Y	Y	Y
IND/YEAR	Y	Y	Y	Y	Y
R^2	0.413	0.525	0.471	0.525	0.453

5.2. Executive political connections and industry nature

This study aims to evaluate the impact of political factions on the disclosure of carbon information related to green technology innovation in corporate leadership. A binary variable is defined to find out whether any economic unit is associated with politics using previous affiliations. In summary, it was clear that natural companies and non-concerned companies were made to undergo regression analyses. Regression analyses are performed for politically linked companies and non-connected enterprises, with the findings shown in Table 9, columns (1) and (2).

Researchers discovered that companies in political associations have a bigger influence with the CID on the GINV of members than their counterparts without political affiliations. The rise in this influence is a function of the power and policy benefits tied to political adherence. Companies with political backing are often entitled to policies that are beneficial, along with monetary incentives, thus facilitating a fast-paced regulatory approval process, which in turn, will promote technical collaboration, bolster market standing and enhance financing options for green projects. As these advantages and chances of getting the finance promote the company's welfare, it means that the financial and logistics challenges are also dropping, and the companies are better prepared to reveal their innovation strategies to the public that are full of the carbon laser theam. A company's CID has a more powerful impact on the viability of green technologies in firms with political links, given that those companies can use the government's push for such sustainable innovations to fast-track their own.

This study also investigated the sectoral differences in the impact of CID on green technology innovation. Two industrial categories, highly polluting and less polluting enterprises, are divided into two groups the samples are tested with the help of different regression models. The research exhibits a strong positive link between corporate environmental transparency and corporate financial performance. At first sing, the most logical assumption would be that those firms with lower levels of pollution have better pollution to performance correlations. The reason can be organizations with lower pollution levels often show higher sustainability and improve the quality of carbon disclosure, which can result in the positive correlation between CID and GINV. A more open environmental report is another way they could get a good reputation. This is because it lets readers know where the company stands in terms of pollution control over time The better quality of carbon disclosure will make those enterprises that are doing it to not only achieve good corporate reputation but compete effectively in the green economy marketplace. As a result, the businesses will certainly get more customers and investors who are concerned about environmental issues. However, companies with high levels of pollution tend to be exposed to more regulations and heavily criticized publicly, which might cause them to pay less attention to the quality of carbon-dioxide disclosure as they find feathers in symbolic compliance instead. Their main concern is regulatory compliance first and green innovation second, which means the impact of CID and GINV is less, thus the correlation between CID and GINV is down in case of some enterprises.

This study explores the diverse impacts of CID on GINV by categorizing enterprises into high-tech and non-high-tech industries and conducting separate regression analyses for each group. As indicated in Table 9, columns (5) and (6), the results reveal that the positive influence of CID on green innovation is more significant in high-tech firms compared to those in traditional industries. The caliber of CID is crucial in the innovation ecosystem, as it enables effective information transmission, promotes technological cooperation, and improves synergistic research initiatives. This is especially important in high-tech sectors, as companies depend significantly on inter-organizational collaborations and knowledge-sharing systems to foster innovation. In these sectors, transparent and high-quality carbon disclosure functions as a strategic instrument, allowing enterprises to establish credibility, recruit external partnerships, and incorporate sustainability factors into their innovation frameworks. As a result, high-tech companies prioritize carbon transparency, acknowledging its significance in enhancing collaborative creative networks and ensuring sustained technical progress. In contrast, companies in conventional sectors may have a more passive stance toward carbon disclosure. Owing to diminished competitive constraints for sustainability-oriented innovation, these enterprises may not emphasize the enhancement of CID quality to the same degree as their high-tech rivals. Instead of using carbon transparency to promote green innovation, conventional industrial enterprises may prioritize compliance with regulatory mandates above the active incorporation of carbon disclosure into their innovation plans.

6. Conclusion and policy recommendations

This study is a comprehensive estimate on the mechanics of corporate carbon disclosure on green technology innovation, focusing on A-share listed firms on the Shanghai and Shenzhen stock exchanges from 2013 to 2020. The study involves an application of textual analysis and the entropy weight approach to elaborate a corporate carbon disclosure index related to the key phrases findings in ESG reports of the company. The conclusions in the research directly show the central dynamics via which the corporate carbon information disclosure affects green technology innovation and the numerous profound implications. The research findings reveal that the carbon information disclosure robustness of a company largely drives corporate green technology innovation, use of funds being relatively a partial mediator in this process. Firms whose carbon footprint transparency is enhanced face a reduced number of financial barriers, thus becoming a funky part of green technology innovations.

An investigation of mediation possibilities shows that, for the most part, the government's financial remuneration as well as the media's coverage primarily influence the later stage of this process. To be more precise, in light of the fact that the government support allocation and media surveillance are, respectively, the main mitigating factors for financing, the beneficial outcomes of CID for green technology innovation are thereby reinforced. This emphasizes the important role played by the external incentives in the promotion of sustainable company growth. The results are robustly validated by employing multiple approaches like taking a two-period delayed explanatory variable as an IV,

Table 9
Heterogeneity Test:Executive political connections and industry nature.

	(1)	(2)	(3)	(4)	(5)	(6)
	PC	N-PC	LP	HP	H-TEC	L-TEC
CID	0.5509**	0.5674	0.5505**	0.4357**	0.8732***	0.2059
	(0.2716)	(0.1687)	(0.1745)	(0.2185)	(0.2199)	(0.1686)
_cons	1.0964**	0.6286**	0.5595**	0.7741*	1.0954**	0.3283
	(0.4777)	(0.2716)	(0.2616)	(0.4133)	(0.3679)	(0.2790)
CONTRONLS	Y	Y	Y	Y	Y	Y
IND/YEAR	Y	Y	Y	Y	Y	Y
R^2	0.544	0.525	0.450	0.528	0.471	0.484

substitute indicators for mediating variables, and using various measures of the dependent variable. The unity of the outcomes through numerous criteria provides support to the reliability of the inquired about issue.

The heterogeneity analysis indicates substantial differences in the influence of CID on green innovation based on varying business ownership structures, geographical regions, and sector attributes. The results demonstrate that the beneficial impact of CID on green innovation is particularly significant in NSOEs and companies situated in central China. This indicates that companies with little financial resources or those situated in areas experiencing industrial change get more advantages from improved carbon transparency. The analysis also reveals significant disparities in the impact of political connections and industry characteristics on the efficacy of CID in promoting green innovation. Firms with political connections, high-tech businesses, and non-heavy polluting sectors have a more robust green innovation response to CID, possibly attributable to their enhanced access to external resources, market incentives, and potential for technical cooperation.

For investors, corporate leaders, and government officials, these findings have major policy ramifications. Policymakers should increase carbon disclosure requirements and support voluntary corporate transparency as better disclosures increase financial availability and encourage sustainable innovation. Second, key mechanisms to remove financial barriers and encourage corporate engagement in green research and development should be government subsidies and media control. Different regulatory frameworks must ultimately be used to address the geographical and ownership-related disparities in CID effectiveness, thereby ensuring that businesses with different structural characteristics may fully employ carbon transparency to support green technological growth.

Based on the research data, this study has indicated a number of policy recommendations for improving the carbon emissions disclosure of companies and also for the development of green technology:

Strategy and Carbon Information Disclosure: Businesses must give top priority to the clarity, accuracy, and verifiability of carbon disclosures if they are to increase responsibility and transparency. Both internal and external stakeholders need high-quality carbon data to evaluate a company's capability and dedication to the development of green technology. By means of sustainable and responsible carbon disclosure, one may attract public attention and investors to sustainability projects, hence creating demand for environmentally friendly products and supporting green technical progress. Companies have to take strict control over their carbon management systems and set thorough emissions reporting policies if they are to reach this. These steps will enable businesses to regularly evaluate their environmental impact, pinpoint areas needing work, and help the green technology industry to flourish.

Governmental Support **for Green Technology:** In the field of green technology innovation, the government plays a crucial role especially in terms of financial incentives and policy support, which might result in the development of fresh, reasonably priced, sustainable, and efficient technologies available to a larger audience. For businesses involved in green technology research and development, subsidies provide a great chance to significantly lower expenses, thereby improving their position in the change toward a low-carbon economy. Furthermore, easing businesses' access to money via preferred loans or lowered interest rates helps to remove the financial constraints faced by companies practicing green innovation. Policymakers should enforce carbon reporting standards more stringently, namely requiring more precise and thorough emissions declarations from companies receiving government subsidies. Implementing this will enable the integration of private sector green energy into the market, resulting in cost-effective, precise, and dependable carbon accounting, along with rapid adoption.

The Media's Role in Environmental Accountability: Media coverage of large corporations' participation in environmentally sustainable initiatives significantly influences public response and concurrently alters companies' financial engagement. The media, by performing its obligations regarding environmental performance and carbon disclosure, stimulates the need for enhanced sustainability pledges from corporations. Simultaneously, acting as an autonomous watchdog, examining the media ensures the completeness of the organization's carbon disclosure and enhances sustainable environmental stewardship. Enhancing the reputation of a company involved in corporate social responsibility on media platforms has the remarkable ability to elevate the firm's prestige, attract investment in sustainable technology, and perhaps facilitate significant advancements in the creation of environmentally friendly goods for the future.

7. Research limitations and perspectives

First, the limitation of the research object is that this study is mainly based on the data of Chinese A-share listed companies, so the findings may be limited by China's economic, policy, and market environments, and it is difficult to directly generalize them to firms in deregulated, market-dominated economies or other countries. Future studies may expand to listed companies in other countries or expand the sample to unlisted companies in China to further verify the generalizability of the findings.

Second, the limitation of data sources is reflected in the fact that this study mainly relies on secondary data such as corporate annual reports, ESG reports, Wind database and CNRDS database, etc. Although these data are widely used in academic research, they may still suffer from incomplete disclosure of information, inconsistent industry standards, or data lag, which may lead to measurement errors. Future studies can combine corporate research, questionnaire surveys or high-frequency environmental monitoring data to improve the accuracy of the data and the reliability of the findings.

Finally, the limitation of the heterogeneity analysis is that although this study has explored the effects of firm ownership structure, regional differences and industry characteristics, it has not yet analyzed in depth potential factors such as the level of corporate governance, international market dependence, and the degree of supply chain greening, etc. Future studies can further combine individual firm characteristics, market competition intensity, or multinational firms' environmental performance comparisons in order to broaden the scope of application of the study.

CRediT authorship contribution statement

Shi Zheng: Project administration, Formal analysis, Data curation. **Chao Wang:** Writing – original draft, Validation, Resources, Data curation. **Ahmad Nawaz:** Writing – review & editing, Resources, Investigation, Conceptualization. **Shamrez Ali:** Writing – review & editing, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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